

Dushan Kavishka

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EDUCATION

Master of Science in Coastal and Port Engineering

Aug 2025 – May 2027

University of North Florida, Jacksonville, FL — GPA: 4.00/4.00

- Thesis: Hurricane-Driven Storm Surge and Barrier Island Morphodynamic Modeling at Matanzas Inlet, FL.
- Relevant Coursework: Computational Fluid Mechanics, Floating Structures, Coastal Structures, Numerical Modeling, Aquatic Chemical Processes, Coastal Estuarine Processes.

Bachelor of Science in Civil Engineering (Honors)

Dec 2019 – Jul 2024

University of Moratuwa, Colombo, Sri Lanka — GPA: 3.45/4.00 — Final Semester: 3.97/4.00

Washington Accord Accredited

- Relevant Coursework: Fluid Mechanics, Hydraulic Engineering, Hydraulic Design, Coastal and Port Engineering, Environmental Engineering, Surveying, Applied Statistics, Irrigation Engineering.

TECHNICAL SKILLS

Numerical Modeling: Delft3D (FLOW-WAVE coupled, SWAN), XBeach, HEC-HMS, ANSYS Fluent, MIKE

GIS & CAD: ArcGIS Pro, AutoCAD, Civil 3D, Revit

Programming & Other Tools: MATLAB, Python, Fortran, Microsoft Office, Microsoft Project

WORK EXPERIENCE

Graduate Research Assistant and Teaching Assistant

Aug 2025 – Present

University of North Florida, Jacksonville, FL

- Developed and calibrated a coupled Delft3D FLOW-WAVE numerical model for storm surge, wave transformation, and sediment transport modeling at a Florida barrier island under hurricane conditions; validated against NOAA tide gauge observations, achieving a smaller RMSE.
- Processed BlueTopo bathymetric datasets and performed geospatial analysis in ArcGIS Pro for nested-grid domain setup; used MATLAB and Python for large-scale NOAA dataset processing and simulation output visualization.
- Applied TPXO10 tidal constituents and NOAA RAP atmospheric forcing for tidal and hurricane event simulations of Hurricanes Matthew (2016) and Dorian (2019).
- Served as Teaching Assistant for Numerical Methods and Computing, independently delivering lectures on root-finding algorithms and numerical methods to undergraduate Civil Engineering students.

Research Engineer — Coastal and Water Resources

Oct 2024 – Aug 2025

Lanka Hydraulic Institute Ltd (LHI), Colombo, Sri Lanka

- Led physical modeling tests (2D flume and 3D wave basin) for three major fishery harbor projects in Oman (Lakbi, Hadbin, Haramel), evaluating wave loading and harbor calmness for coastal structure design.
- Conducted numerical wave propagation and coastal hydrodynamic simulations using MIKE; analyzed wave transformation, diffraction, and reflection patterns for port infrastructure design.
- Performed hydrographic field surveys and geotechnical investigations for coastal and port projects including ASYAD Drydock at Duqm Port; prepared technical reports and presented findings to clients.

Civil Engineering Intern

Jan 2023 – May 2023

Maga Engineering (Pvt) Ltd, Colombo, Sri Lanka

- Gained practical experience in construction project planning, field reconnaissance, site inspection, and on-site safety supervision across active civil infrastructure projects.

PROJECTS AND PUBLICATIONS

- **Oral Conference Presentations** — Master's research presented at ICCE 2026 (Galveston, TX) and the 2026 UF Water Institute Symposium (Gainesville, FL); covering hurricane storm surge modeling, barrier island breaching, and coupled wave-hydrodynamic numerical modeling.
- **Springer Nature Publication** — Won Best Scientific Paper Award at ICSBE 2024; numerical investigation of wind loading on gable-roof low-rise buildings using ANSYS Fluent CFD simulations (DOI: 10.1007/978-981-96-8301-7_21).
- **Undergraduate Design Project** — Planning and design of a coastal resort including Environmental Impact Assessment (EIA) and flood hydrologic modeling using HEC-HMS.
- A journal publication is planned based on the Graduate Research Assistant work and the master's thesis.